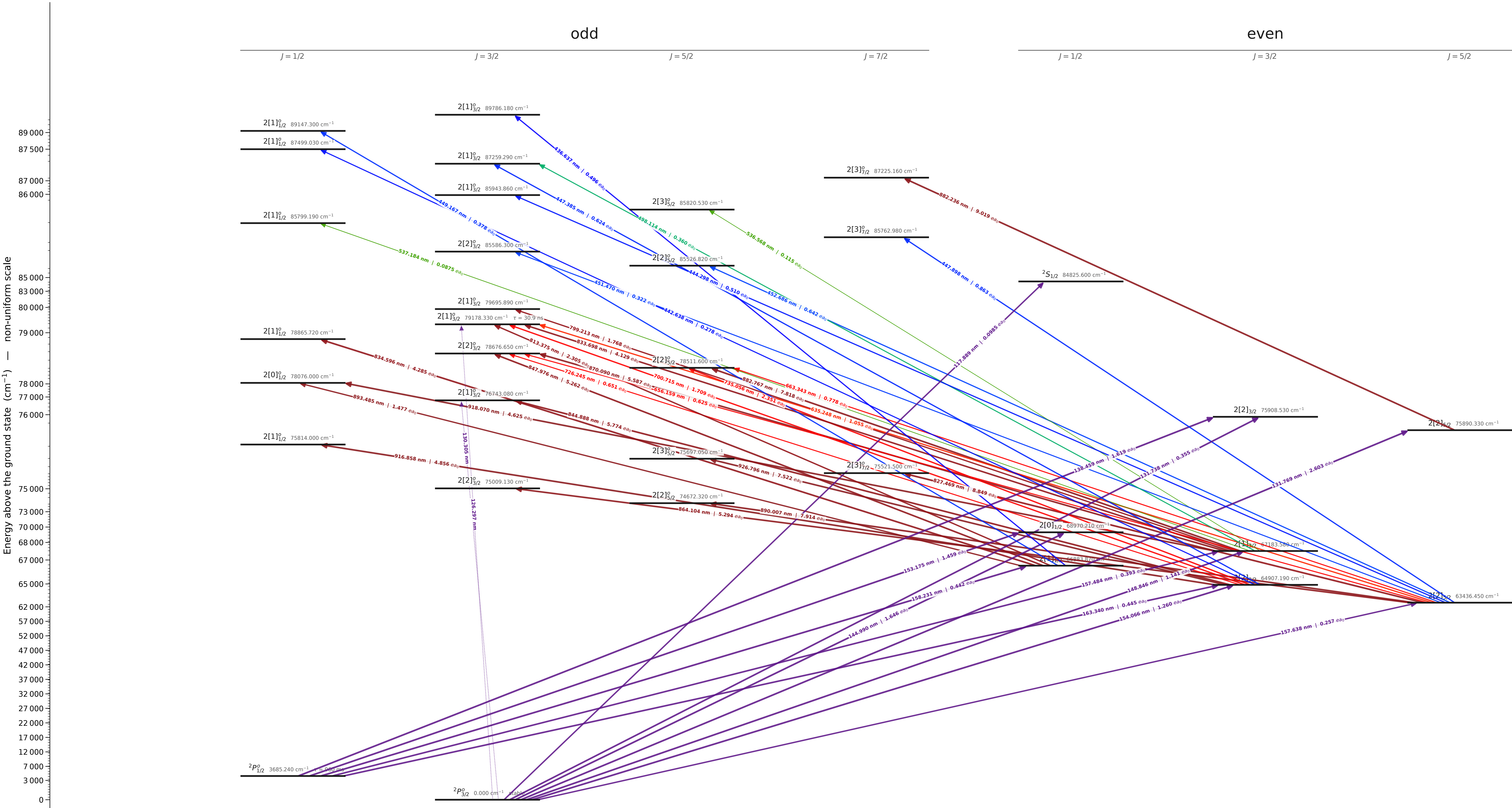
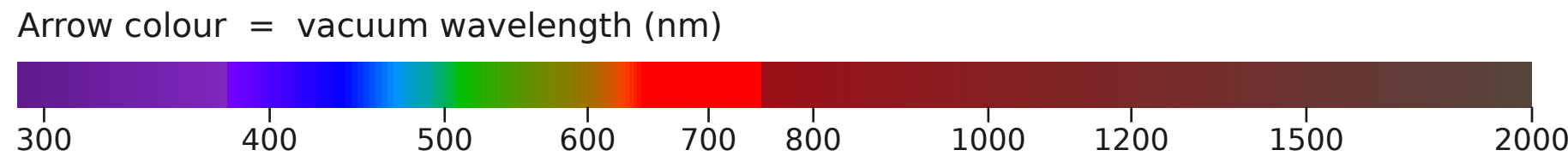


# Bromine

Br (Br I). Strongest classified transitions below 2  $\mu\text{m}$  (NIST): 33 levels, 47 lines.

Br<sup>+</sup> ionisation limit 95284.80 cm<sup>-1</sup> = 11.81381 eV (104.949 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $|Q||er||J\rangle\rangle|$  in  $ea_0$   
 Matrix elements are measured or NIST values; \* = high-accuracy theory,  $\approx$  = model calculation.  
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $ \langle J  \text{er}  J' \rangle $ (ea) | A (s <sup>-1</sup> ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Use                                                          |
|---------------------------|--------------------|--------------------|-------------|-------------------------------------------|----------------------|-------------------------|------------|--------------------------------------------------------------|
| $2^2P_{3/2} - 2[1]_{3/2}$ | 126.2972           | -                  | 2373.7066   | -                                         | -                    | 5.151                   | -          | TALIF detection of ground-state Br: 2 x 252.594 nm, fluor... |
| $2^2P_{3/2} - 2[1]_{3/2}$ | 130.3049           | -                  | 2300.6997   | -                                         | -                    | -                       | -          | two-photon LIF scheme near 260.6 nm with fluorescence 5p...  |
| $2^2P_{3/2} - 2[1]_{3/2}$ | 148.8459           | -                  | 2014.1131   | 1.141                                     | 2.000e+08            | -                       | -          | 148.8 nm resonance line                                      |
| $2^2P_{3/2} - 2[2]_{3/2}$ | 154.0661           | -                  | 1945.8686   | 1.260                                     | 2.000e+08            | -                       | -          | 154.1 nm resonance line                                      |
| $2^2P_{1/2} - 2[1]_{3/2}$ | 157.4844           | -                  | 1903.6323   | 0.393                                     | 2.000e+07            | -                       | -          | 157.5 nm line to the metastable $2^2P^{1/2}$ level           |
| $2^2P_{3/2} - 2[2]_{5/2}$ | 157.6381           | -                  | 1901.7769   | 0.257                                     | 5.700e+06            | -                       | -          | 157.6 nm resonance line                                      |
| $2^2P_{1/2} - 2[2]_{3/2}$ | 163.3401           | -                  | 1835.3879   | 0.445                                     | 2.300e+07            | -                       | -          | 163.3 nm line to the metastable $2^2P^{1/2}$ level           |
| $2[2]_{5/2} - 2[1]_{3/2}$ | 635.2481           | 635.0725           | 471.9297    | 1.055                                     | 2.000e+06            | 5.151                   | 6.8        | 635.25 nm fluorescence line used to detect Br atoms after... |

$\lambda$ ,  $\nu$  from NIST level energies unless a measured frequency for this isotope is available.  $A$ : measured value or NIST.  $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ .

### <sup>79</sup>Br hyperfine structure (*I* = 3/2)

| Level      | A (MHz)      | B (MHz)        | Shift of each F level from the centre of gravity (MHz) |               |              |               |
|------------|--------------|----------------|--------------------------------------------------------|---------------|--------------|---------------|
| $2p_{3/2}$ | 884.80977(6) | -384.88284(20) | F=0: -3799.14                                          | F=1: -2529.45 | F=2: -374.94 | F=3: +1894.60 |
| $2p_{1/2}$ | 5332.55(4)   | -              | F=1: -6665.69                                          | F=2: +3999.41 |              |               |

Measured A, B constants; sources in the data file.